
OTRec: A Deep Learning Recommender for Druggable Disease–Target Prioritization

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Identifying druggable disease–target associations remains a central challenge in
2 translational medicine, limiting therapeutic discovery and repurposing. Here, we
3 present OTRec, a deep learning–based recommender system that ranks such associ-
4 ations at scale and evaluates them in a temporal hold-out setting. Unlike approaches
5 that rely on manually curated or aggregated evidence scores, OTRec employs a
6 two-tower architecture to learn latent representations from 663,351 disease–target
7 pairs. The model integrates heterogeneous inputs, including textual descriptions,
8 ontology-derived features, and biological annotations such as tractability, Gene
9 Ontology (GO) terms, and pathway information. We perform temporal validation
10 by training on the 2022 Open Targets release and evaluating on clinical trial data
11 from 2025. OTRec improves on the retrospective Open Targets association score on
12 ROC-AUC (0.872 ± 0.005 vs. 0.559); on PR-AUC it remains close to a target-prior
13 baseline (0.288 ± 0.009 vs. 0.299 , gap 0.011). Using 5×5 target-disjoint cross-
14 validation, OTRec reaches ROC-AUC 0.950 (PR-AUC 0.844), narrowly improving
15 on the same-feature CatBoost baseline. At inference we rank the druggable genome
16 across $\sim 19,000$ OTP diseases and release $\sim 282,500$ candidate associations above
17 a 0.65 score threshold (in-distribution CV precision 0.92 ; out-of-distribution pre-
18 cision not directly estimated), covering $4,346$ diseases including $2,322$ orphan
19 diseases, through an interactive prediction platform. Code, weights, predictions.
20 The public demo is publicly available and will be released after the review period.

21 1 Introduction

22 Drug discovery is hindered by a pervasive “streetlight effect”: research concentrates on well-studied
23 targets while large regions of the druggable genome remain under-investigated [3, 22, 9]. The
24 Open Targets Platform (OTP) integrates diverse evidence sources, from GWAS associations to text-
25 mined literature, but the resulting association scores are intrinsically retrospective, reflecting existing
26 knowledge rather than a target’s prospective therapeutic promise. A zero or low score may signal
27 the absence of investigation rather than a genuine lack of relevance. Moreover, manually designed,
28 rule-based scoring frameworks typically underperform relative to data-driven, learned approaches [2].

29 We therefore developed **OTRec** (Open Targets Recommender). The task is: given a disease, rank
30 every gene in the druggable genome by its probability of entering clinical trials, a label OTP captures
31 across $>17,000$ diseases. In this work, “target” denotes an OTP gene target (mapped to Ensembl
32 gene IDs), and “drug” denotes therapeutic evidence represented in OTP via approved or clinical-
33 stage associations linked to those targets. Unlike OTP’s rule-based aggregations [8], OTRec uses a
34 two-tower neural architecture that separately encodes disease and target representations from raw
35 biomedical text and ontology. These representations are scored by cosine similarity, which natively
36 enables attribute-cold-start inference for targets absent from training folds but still represented through

public annotations. Validated in a temporal hold-out against later trial entries, it improves on the retrospective OTP score and the learned baselines in the target-disjoint CV, and we release candidate associations $\sim 282,500$ candidate associations above a 0.65 score threshold with a fully interactive platform.

Our contributions are threefold: (i) A two-tower neural recommender that learns disease and target representations from raw biomedical text and ontology, supporting cold-start prediction for novel, annotated targets. (ii) Provide a temporal-hold-out benchmark that combines target-disjoint CV with a 2022 $\rightarrow$ 2025 temporal split on the OTP target-prioritization task, on which OTRec improves on the retrospective OTP score while remaining competitive with OTP-native tree baselines. (iii) A public, ranked candidate dataset of $\sim 282,500$ novel disease–target associations covering 4,346 diseases (including 2,322 orphan diseases) with an interactive query interface.

2 Related Work

Disease–target prioritization sits between recommendation, biomedical knowledge integration, and drug–target interaction (DTI) prediction. We organize prior work into four threads and explain why none directly address prospective target prioritization on OTP.

OTP-native ML. The closest prior work, Han et al. [8], trains gradient boosting on 14 OTP evidence-channel features and is the strongest publicly reimplementable baseline we can include in our benchmark. It inherits the retrospective bias of OTP evidence channels, which is precisely what our temporal evaluation is designed to expose.

Knowledge-graph and graph neural network DTI. KG embedding and GNN methods, such as DRKG and KGNN-LS, integrate heterogeneous biomedical entities and relations [21, 25, 16]. These methods rely on a pre-curated interaction graph whose edges themselves encode the same investigation bias we aim to escape. Furthermore, they are typically evaluated on curated DTI benchmarks (BindingDB, DAVIS) with balanced positive/negative ratios, and do not natively support the OTP implicit-feedback, target-disjoint, time-split regime. We include Matrix-Factorization and TF-IDF controls that serve as comparable cold-start relational baselines.

Chemistry-centric DTI. Deep DTI models such as DeepPurpose and MolTrans and related transformer approaches [10, 12] predict binding affinity from compound structure and protein sequence. Their inputs (SMILES, sequence) and labels (binding affinity) are orthogonal to the disease-level clinical relevance signal modeled here, so they cannot be evaluated on the same benchmark without redefining the task.

Biomedical text encoders and prospective benchmarks. Biomedical language models (BioBERT, PubMedBERT [7]) and benchmarking suites such as Therapeutics Data Commons [11] provide reusable text representations and standardized splits, respectively. Neither defines a disease–target prioritization task with prospective clinical-trial-entry labels at OTP scale; we introduce this benchmark and release splits, code, and candidate predictions to enable direct replication and comparison.

3 Methods

3.1 Problem Formulation and Data

We formulate target prioritization as an implicit-feedback recommendation task: rank candidate human protein-coding genes for a given disease by likelihood of clinical relevance. We used Open Targets Platform (OTP) Release 25.06 for cross-validation and inference, and Release 22.02 as the historical training snapshot for temporal validation [3, 20, 18, 19]. Positive labels ($y=1$) are disease–target pairs with evidence of clinical progression (clinical trials or approved drugs) in the relevant OTP known-drug / clinical-evidence table; all constructed rows without such evidence are implicit negatives ($y=0$). The implicit-feedback formulation conflates true negatives with unlabeled positives; we treat this label noise as inherent to the prospective task. The temporal split (§3.3) re-labels pairs whose clinical evidence appears only in a later OTP release. Operationally, we construct the temporal test set by an anti-join: retain every 2025.06 row whose exact (diseaseId, targetId, y) tuple is absent from the processed 2022.02 frame. This keeps 2022-negative / 2025-positive conversions, excludes pairs already positive in 2022, and retains all 2025-negative rows that enter the processed 2025 frame under the same rule. Negatives are exhaustive within this anti-join cohort rather than sampled.

No special censoring is applied for diseases or targets with different clinical-trial ascertainment, prevalence, or commercial interest; these biases remain part of the temporal-labeling problem. The OTP known-drug exports used here do not a consistently available trial-start-date field, so we do not independently censor by ClinicalTrials.gov start date. The resulting test set covers 11,624 diseases (vs. 9,741 diseases in the 2022 training frame), making it strongly covariate-shifted in both disease coverage and label distribution.

Scope of “drug” and “gene”. Throughout, “drug” follows OTP usage and covers any therapeutic entity in the OTP drug index ($N=18,081$ entities in Release 2025.06, of which 10,956 have reached clinical phase ≥ 1): small molecules ($N=14,848$, 82% of the index), antibodies ($N=963$), proteins / peptides / enzymes ($N=830$), oligonucleotides ($N=159$), antibody–drug conjugates ($N=119$), gene therapies ($N=117$), cell therapies ($N=52$), and vaccines. “Gene” / “target” refers to the human protein-coding subset of the Finan et al. druggable genome [6] ($\sim 4,600$ genes, of 20,130 protein-coding genes in OTP). Non-coding loci catalogued by OTP (lncRNAs $N=34,882$, miRNAs $N=1,879$, snRNAs, pseudogenes) are excluded from both the training label set and the inference pool, mirroring the small-molecule / biologic-modality bias of the OTP drug index; we revisit this scope in §5.

We applied filtering criteria aligned with prior work [8]: non-specific ontology terms (“measurement”, “phenotype”, “biological process”, “cell proliferation disorder”) were excluded. Training restricts to $\sim 1,500$ targets with clinical associations across any indication, excluding targets with only a single-disease evidence record to reduce disease-specific overfitting. The training set comprises 663,351 disease–target pairs spanning $\sim 6,000$ positive-bearing diseases (out of $>12,000$ diseases overall in the training pool; 67,532 positives, 10%). During inference, the model ranks the full druggable genome [6] ($\sim 4,600$ genes) across $\sim 19,000$ valid OTP diseases. Note that CV metrics characterise generalization within the multi-indication-clinical training subset and should be read as an upper bound on full-druggable-genome inference quality.

3.2 Model Architecture and Features

OTRec uses a multi-input two-tower architecture (Figure 1A) that separately encodes disease and target representations, enabling the model to learn semantic and biological similarities from text and ontology structure.

Inputs. Text features are processed via Text Vectorization (bag-of-words, count mode). The *disease tower* uses disease names, descriptions, exact synonyms, therapeutic areas, phenotypes, and ontology parents (EFO terms [14]), concatenated with a learned disease-ID embedding ($d=64$). The *target tower* uses gene symbol, name, synonyms, function descriptions, Gene Ontology terms [23], pathway annotations including Reactome-derived terms [15], UniProt-derived annotations [24], and OTP target prioritisation / tractability features including gnomAD-derived genetic constraint scores where available [20, 13].

Architecture and training. Each tower applies a Residual “Wide & Deep” block: a direct linear projection ($d=384$) summed with a nonlinear bottleneck path (Dense $384 \rightarrow 64 \rightarrow 384$, ELU activations, Dropout 0.15–0.35). The 384-dimensional embeddings interact via normalized dot product (cosine similarity), followed by a multi-task head: a sigmoid output predicts clinical trial entry while an auxiliary linear head predicts the OTP overall association score from the same release used to construct the training split (2025.06 in target-disjoint CV, 2022.02 in the temporal experiment). In the temporal experiment, the auxiliary target is the 2022.02 OTP overall score, so the model does not use post-cutoff information. Training used Adam ($LR 8 \times 10^{-3}$, batch size 1,024) with dynamic LR annealing and early stopping on a held-out 5% validation set. Implementation: TensorFlow/Keras [4].

3.3 Evaluation and Baselines

Target-disjoint CV. To evaluate generalization to novel targets unseen during training, we use 5×5 -fold target-disjoint cross-validation (5 folds, 5 repeats, 25 folds total). In each fold, 20% of targets are withheld; all associations involving held-out targets are removed from training. This is attribute-cold-start rather than annotation-free cold-start: held-out targets remain represented by public text, ontology, pathway, and tractability features.

Temporal split. A model trained on OTP Release 2022.02 is tested on pairs with no clinical evidence in 2022 that entered clinical trials by 2025. The test set covers $\sim 12,000$ diseases (vs. $\sim 6,000$ positive-bearing diseases at training time), making it strongly covariate-shifted in both disease coverage and label distribution. Temporal labels are defined by release difference: we construct the 2025 cohort and remove any exact (diseaseId, targetId, label) tuple already present in 2022.

Baselines. We benchmark against six primary methods spanning ML, ontology-aware, and cold-start controls: (i) **OTTree**: CatBoost [5] trained on the same OTRec feature set, isolating the contribution of the two-tower architecture from the feature engineering; (ii) **Han et al.** [8]: XGBoost on the 14 OTP evidence-channel scores (genetic association, somatic mutation, known drug, affected pathway, RNA expression, literature, animal models, etc.), the strongest publicly reimplementable ML baseline for the OTP target-prioritization task; (iii) **Open Targets Score**: the raw OTP overall association score (which itself aggregates clinical evidence and is therefore an upper bound on retrospective performance); (iv) **Disease Mean**: the per-disease positive rate computed on the training split, broadcast to every candidate target for that disease (a strong historical-frequency prior in the target-disjoint setting); (v) **Target Mean**: the per-target positive rate computed on the training split, broadcast to every candidate disease for that target; (vi) **Matrix Factorization**: TruncatedSVD on the binary positive interaction matrix; (vii) **TF-IDF cosine**: lexical similarity between concatenated disease and target text fields. These simple controls verify that the target-disjoint regime is non-trivial. In particular, a leakage-free Target Mean is expected to collapse toward the global prior because test targets are unseen in every fold. We also include **Disease Mean** in the temporal table as a disease-prior control.

4 Results

4.1 Benchmark Comparison

In the 5×5 target-disjoint CV (Table 1), OTRec achieves ROC-AUC 0.950 (PR-AUC 0.844), marginally surpassing OTTree (0.947 / 0.772) and exceeding the Open Targets Score (0.914 / 0.455). Because Han et al. uses only 14 OTP evidence channels whereas OTRec and OTTree use the richer text+ontology+tractability feature set, OTRec’s gain over Han et al. reflects both feature engineering and model class. The OTRec–OTTree gap isolates the two-tower contribution on matched inputs (+0.003 ROC-AUC). Among the simple controls, Disease Mean is the strongest historical prior (0.873 / 0.466), whereas a leakage-free Target Mean collapses to the global prior (0.500 / 0.102) because every test target is unseen in its evaluation fold. Matrix Factorization (0.811 / 0.265) and TF-IDF (0.513 / 0.106) remain much weaker than OTRec, indicating that the target-disjoint regime genuinely tests generalization to novel targets and that OTRec’s advantage comes from learned biological content transfer rather than interaction memorization or surface text matching.

In the temporal setting (Table 2), OTRec improves over the retrospective OTP score on ROC-AUC by 0.314 (0.872 ± 0.005 vs. 0.559), while achieving PR-AUC (0.288 ± 0.009) close to the stronger historical Target Mean prior (0.299, gap 0.011) and well above a disease-only prior (0.052). Fast cold-start controls are substantially weaker in this prospective regime (Matrix Factorization: 0.553 / 0.050; TF-IDF cosine: 0.512 / 0.042), indicating that historical priors and learned representations carry most of the forward signal. Historical target-level priors carry real prospective signal; OTRec’s advantage over them is concentrated in pair-level ranking (ROC-AUC), not in simply recovering which targets already accumulated many known indications. The temporal model uses the 2022 OTP overall score for both its auxiliary target and the OTP-score baseline.

4.2 Disease-Centric Shortlist Utility

Aggregate ROC-AUC and PR-AUC measure pairwise discrimination, but operational target prioritization is shortlist-driven: a curator typically inspects only the first few predictions per disease. We re-analysed target-disjoint out-of-fold predictions disease-by-disease, restricting to the 2,329 diseases with at least one held-out positive target (Table 3).

OTRec placed a true positive at rank 1 for 68.2% of diseases (vs. 48.7% for OTTree) and improved mean reciprocal rank (0.741 vs. 0.600). At rank 5, the advantage remained (80.6% vs. 73.2%); at rank 10 the two methods tied (84.8% each), and OTTree won at rank 25 (95.1% vs. 89.7%). At the same shortlist depth, OTRec also improved mean precision@5 (64.5% vs. 53.7%) and mean



Figure 1: **OTRec architecture and performance.** (A) The two-tower model separately encodes disease and target features (text, ontology, tractability, gene symbol/name) into 384-dimensional embeddings compared by normalized dot product (cosine similarity) to predict clinical trial entry probability. (B) Target-disjoint 5×5 cross-validation results. OTRec achieves the highest mean ROC-AUC (0.95) and PR-AUC (0.844); error bars denote standard deviation. (C) Temporal hold-out validation (training on 2022, predicting 2025 entries). Panel C shows mean performance over five random initialization seeds; error bars denote standard deviation.

191 recall@5 (32.1% vs. 27.2%). OTRec is preferable for top-1 / top-5 curator workflows that surface a
 192 short actionable shortlist; OTTree is preferable for broader screens at $k \geq 25$ where its more diffuse
 193 predictive distribution recovers more positives.

194 4.3 Disease Repurposing Candidates

195 We identified $\sim 282,500$ candidate associations above a 0.65 score threshold for 4,346 diseases
 196 matched with 2,808 targets, including 2,322 orphan diseases, where “orphan” here refers to OTP dis-
 197 eases with no clinically associated target in 2025.06 rather than a formal prevalence-based regulatory
 198 definition. Candidates were generated by screening the druggable genome, removing known links,
 199 and retaining predictions with score > 0.65 —a threshold selected by inspecting the precision-recall

Table 1: Benchmark performance in 5×5 target-disjoint cross-validation (mean $\pm$ SD over 25 folds). A horizontal rule separates ML baselines from cold-start controls.

Model	ROC-AUC	$\pm$ SD	PR-AUC	$\pm$ SD
OTRec (ours)	0.950	0.007	0.844	0.017
OTTree (CatBoost)	0.947	0.005	0.772	0.023
Han et al. (2022)	0.910	0.014	0.726	0.046
Open Targets Score	0.914	0.005	0.455	0.030
Disease Mean	0.873	0.004	0.466	0.022
Target Mean	0.500	0	0.102	0.008
Matrix Factorization	0.811	0.008	0.265	0.014
TF-IDF cosine	0.513	0.016	0.106	0.009

Table 2: Temporal hold-out validation: trained on OTP 22.02, tested on pairs whose 2025.06 label was retained by the temporal anti-join (mean $\pm$ SD over 5 random seeds).

Model	ROC-AUC	$\pm$ SD	PR-AUC	$\pm$ SD
OTRec (ours)	0.872	0.005	0.288	0.009
OTTree (CatBoost)	0.855	0.004	0.223	0.005
Target Mean	0.832	0	0.299	0
Matrix Factorization	0.553	0	0.050	0
TF-IDF cosine	0.512	0	0.042	0
Disease Mean	0.594	0	0.052	0
Open Targets Score	0.559	0	0.082	0

curve on out-of-fold CV data (92% in-distribution precision, 62% recall; up to 200 per disease). Out-of-distribution precision for genuinely novel pairs is lower and not directly estimated; the ranked prediction set should not be treated as a validated discovery list. Predictions were further ranked and annotated using the InterFeat framework [17]; full candidates are available in the repository (supplementary text S1). The interactive platform filters out known disease–target pairs, so known associations are intentionally absent.

Additionally we release an interactive Gradio web demo [1] (Figure 2) that lets users issue both forward (disease $\rightarrow$ ranked targets) and reverse (target $\rightarrow$ ranked diseases) queries against the trained model, with links to OTP and EFO or ontology records, tractability annotations, comparison labels, and CSV export of the full druggable-genome ranking.

4.4 Discovery Examples

The following examples are illustrative of the candidates the model ranks highly; they are not part of a systematic screen, are not experimentally validated, and are outside the quantitative evaluation. Full ranked lists are released with the dataset.

Temporal hold-out example: *POLA1* / *POLA2* in central nervous system cancer (2022 $\rightarrow$ 2025). OTRec ranked the catalytic and accessory subunits of replicative DNA polymerase α — *POLA1* (score 0.961) and *POLA2* (score 0.966) — among the top candidates for central nervous system cancer (EFO_0000326), despite a near-zero OTP known-drug score in 2022 ($\sim$ 0.06). By 2025 both pairs had clinical-progression, coincident with inhibitors of the POL α /primase complex entering brain-tumour trials. We present this as a biology-first temporal example rather than a ready repurposing claim: recent glioma literature supports replication-stress dependencies and reports preclinical activity of DNA polymerase- α inhibition, but direct POLA1/POLA2 clinical validation remains limited

CDKL5 disorder. OTRec assigned very high scores to *SCN8A* (0.991) and *SCN1A* (0.989) within a broader sodium-channel module in CDKL5 disorder, a rare developmental and epileptic encephalopathy with persistently refractory seizures. This is interesting because multicenter cohort data suggest sodium-channel blockers can benefit a subset of CDKL5 patients, although our model does not identify which channel, or which direction of modulation, is optimal.

Table 3: Disease-centric shortlist utility across 2,329 diseases with ≥ 1 held-out positive target.

Model	Hit@1	Hit@5	P@5	R@5	Hit@10	Hit@25	MRR
OTRec	68.2%	80.6%	64.5%	32.1%	84.8%	89.7%	0.741
OTTree (CatBoost)	48.7%	73.2%	53.7%	27.2%	84.8%	95.1%	0.600

Figure 2: **OTRec interactive demo.** Users select a disease or target; the Gradio app returns the ranked candidate list with OTP/EFO cross-links where supported, tractability category, and a CSV export of the full druggable-genome ranking.

227 **Limited cutaneous systemic sclerosis.** OTRec ranked *PDE4C* highly (0.924) for limited cutaneous
228 systemic sclerosis, a fibro-inflammatory disease with few disease-modifying options. This is a
229 plausible translational case rather than a validation claim: apremilast and other PDE4 inhibitors
230 show anti-fibrotic and anti-inflammatory activity in systemic-sclerosis fibroblast and bleomycin-skin
231 models, making oral PDE4 inhibition a tractable repurposing hypothesis.

232 **Temporal arteritis / giant cell arteritis.** *DHODH* scored 0.829 for temporal arteritis / giant cell
233 arteritis, pointing to a lymphocyte-metabolism axis rather than another cytokine-only mechanism.
234 This is more actionable than many orphan predictions because DHODH inhibition is already clinically
235 accessible through leflunomide, and recent observational studies suggest steroid-sparing benefit in
236 giant cell arteritis.

237 **Spinal muscular atrophy (SMA).** SMA is caused primarily by insufficient SMN protein from
238 *SMN1* loss, and current therapies target SMN1/SMN2 biology. OTRec nevertheless ranked ion-
239 channel and synaptic-signaling candidates, including potassium voltage-gated channel family mem-
240 bers and *GABBR1*. These predictions should be read as hypothesis-generating symptomatic or
241 modifier targets rather than disease-causal replacements for SMN restoration: they point to neuronal
242 excitability and inhibitory signaling as possible complementary axes for model-system follow-up.

243 **Croup and Coxsackievirus.** OTRec ranked FKBP1A, an immunophilin co-opted by viral replica-
244 tion complexes, among the top candidates; further rationale is documented in the repository.

245 **Limited cutaneous systemic sclerosis.** PDE4C was ranked highly as a mechanism-based repurpos-
246 ing candidate; apremilast (a PDE4 inhibitor) is already approved for related indications.

247 **Ulcerative colitis.** OTRec ranked IL6R highly; clinical context for IL6-axis blockade in UC is
248 documented in the repository.

4.5 Feature Contributions and Model Characteristics

Characterization of novel predictions. Novel predicted targets (lacking approved drugs) are less genetically constrained ($P < 3.3 \times 10^{-19}$, Cohen’s $d=-0.37$) and enriched for secreted proteins (+8.4%, $P < 2.1 \times 10^{-7}$), indicating divergent tractability profiles. They map to significantly fewer annotated pathways ($P < 7.2 \times 10^{-50}$), reflecting lower annotation coverage rather than functional irrelevance. Paralog counts and sequence identity do not differ significantly ($P=0.63$), so this analysis does not suggest that OTRec is merely prioritizing homologs of known genes. The model assigns higher confidence to disease-specific targets (median 0.936) vs. multi-disease hubs (median 0.878; $P < 6.3 \times 10^{-17}$), consistent with mechanistic specificity; representative high-degree hubs include GUCY1B2, tubulins, FKBP1A, and DNA maintenance proteins such as POLD2. This pattern is consistent with OTRec ranking pathway- or system-level intervention points shared across diseases rather than only causal-gene recovery, though such candidates still require experimental validation. Comparing out-of-fold prediction scores against the maximum clinical phase (1–4) reached by $\sim 65,000$ known positive associations, we observe a weak but significant positive correlation (Spearman $\rho=0.20$), consistent with OTRec assigning higher scores to candidates that progress to later development stages. All p-values reported in this section are Bonferroni-corrected for the six tests performed.

5 Limitations

Auxiliary supervision and objective dependence. The auxiliary head regresses the OTP overall association score from the same release as each training split (2025.06 in CV; 2022.02 in the temporal run), so the temporal experiment does not use post-cutoff 2025 supervision. The reported temporal mean performance uses a multi-task objective rather than a pure binary objective. An auxiliary-free temporal ablation would isolate how much the score head changes optimization.

Single temporal split period. The 2022→2025 evaluation uses one release pair; while we now report variance across five random initialization seeds, rolling release-pair evaluations (e.g., 2021→2024) would more fully test the stability of the prospective signal and are direct next steps.

Illustrative discovery examples. The four discovery cases are curated to span disease classes; they are not the output of a systematic screen and have not been experimentally validated. The full unfiltered candidate set is released precisely to allow independent, unsupervised evaluation.

Bag-of-words text encoding. We use count-mode text vectorization rather than a biomedical transformer encoder (BioBERT, PubMedBERT). This trades representational richness for inference scale ($\sim 87\text{M}$ pairs at prediction time) and interpretability of contributing tokens. Substituting a frozen biomedical LM into either tower is a natural extension we did not benchmark here.

Implicit-feedback / PU formulation. All non-clinical pairs are treated as $y=0$; no positive-unlabeled correction (Elkan–Noto, nnPU) is applied. The temporal split partially calibrates this by relabeling pairs that entered trials between 2022 and 2025, making the label-noise assumption testable rather than purely assumed.

Druggable-genome inference scope. Inference is restricted to the Finan et al. [6] druggable genome ($\sim 4,600$ protein-coding genes) and to OTP-defined diseases. Non-coding targets (lncRNA, miRNA), intrinsically disordered regions, and rare diseases without OTP records are outside the released candidate set. OTRec also ranks targets independently and does not model multi-target / off-target liabilities; downstream polypharmacology and selectivity filters remain the curator’s responsibility.

6 Conclusion

Ranking all druggable targets across all OTP diseases against future clinical-trial entries is a tractable, precisely defined surrogate for prospective therapeutic relevance, and one where learned representations from text and ontology improve over the retrospective evidence score on ROC-AUC, while PR-AUC matches a strong historical target-prior baseline. OTRec makes this ranking available at scale: $\sim 282,500$ candidate associations, an interactive query interface, and a temporal benchmark against which future models can be directly compared. Rolling release splits, additional data sources, and transformer-based tower encoders are the natural next steps to test whether deeper language representations extend the prospective advantage beyond the BoW ceiling established here.

Availability

During blind review, an anonymized supplementary contains the code needed to reproduce the main analyses and the result CSVs used in the manuscript. The public repository, models, ranked predictions, and interactive query interface are online but not linked during anonymized review; they will be shared publicly after acceptance / camera-ready.

References

- [1] Abubakar Abid, Ali Abdalla, Ali Abid, Dawood Khan, Abdulrahman Alfozan, and James Zou. Gradio: Hassle-free sharing and testing of ml models in the wild, 2019. URL <https://arxiv.org/abs/1906.02569>.
- [2] Qurrat Ul Ain, Antoniya Aleksandrova, Florian D. Roessler, and Pedro J. Ballester. Machine-learning scoring functions to improve structure-based binding affinity prediction and virtual screening. *WIREs Computational Molecular Science*, 5(6):405–424, 2015. doi: 10.1002/wcms.1225. URL <https://doi.org/10.1002/wcms.1225>.
- [3] Annalisa Buniello, Daniel Suveges, Carla Cruz-Castillo, et al. Open targets platform: Facilitating therapeutic hypotheses building in drug discovery. *Nucleic Acids Research*, 53(D1): D1467–D1475, 2025. doi: 10.1093/nar/gkae1128. URL <https://doi.org/10.1093/nar/gkae1128>.
- [4] Francois Chollet. Keras. <https://keras.io>, 2015. Accessed 2026-04-25.
- [5] Anna Veronika Dorogush, Vasily Ershov, and Andrey Gulin. Catboost: Gradient boosting with categorical features support. *arXiv preprint arXiv:1810.11363*, 2018. URL <https://arxiv.org/abs/1810.11363>.
- [6] Chris Finan, Anna Gaulton, Felix A. Kruger, et al. The druggable genome and support for target identification and validation in drug development. *Science Translational Medicine*, 9(383): eaag1166, 2017. doi: 10.1126/scitranslmed.aag1166. URL <https://doi.org/10.1126/scitranslmed.aag1166>.
- [7] Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiang Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23, 2021. doi: 10.1145/3458754. URL <https://doi.org/10.1145/3458754>.
- [8] Yingnan Han, Katherine Klinger, Deepak K. Rajpal, Cheng Zhu, and Erin Teeple. Empowering the discovery of novel target–disease associations via machine learning in the open targets platform. *BMC Bioinformatics*, 23:232, 2022. doi: 10.1186/s12859-022-04753-4. URL <https://doi.org/10.1186/s12859-022-04753-4>.
- [9] Shahar Harel and Kira Radinsky. Accelerating prototype-based drug discovery using conditional diversity networks. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 331–339, 2018. doi: 10.1145/3219819.3219882. URL <https://doi.org/10.1145/3219819.3219882>.
- [10] Kexin Huang, Tianfan Fu, Lucas M. Glass, Marinka Zitnik, Cao Xiao, and Jimeng Sun. Deeppurpose: a deep learning library for drug–target interaction prediction. *Bioinformatics*, 36(22-23):5545–5547, 2020. doi: 10.1093/bioinformatics/btaa1005. URL <https://doi.org/10.1093/bioinformatics/btaa1005>.
- [11] Kexin Huang, Tianfan Fu, Wenhao Gao, Yue Zhao, Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun, and Marinka Zitnik. Therapeutics data commons: Machine learning datasets and tasks for drug discovery and development. In *NeurIPS 2021 Datasets and Benchmarks Track*, 2021. URL <https://openreview.net/forum?id=8nvgN0RnoWr>.
- [12] Kexin Huang, Cao Xiao, Lucas M. Glass, and Jimeng Sun. Moltrans: Molecular interaction transformer for drug–target interaction prediction. *Bioinformatics*, 37(6):830–836, 2021. doi: 10.1093/bioinformatics/btaa880.

- [13] Konrad J. Karczewski, Laurent C. Francioli, Grace Tiao, Beryl B. Cummings, Jessica Alföldi, Qingbo Wang, Ryan L. Collins, Kristen M. Laricchia, Andrea Ganna, Daniel P. Birnbaum, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581:434–443, 2020. doi: 10.1038/s41586-020-2308-7. URL <https://doi.org/10.1038/s41586-020-2308-7>.
- [14] James Malone, Ele Holloway, Tomasz Adamusiak, Misha Kapushesky, Jie Zheng, Nikolay Kolesnikov, Anna Zhukova, Alvis Brazma, and Helen Parkinson. Modeling sample variables with an experimental factor ontology. *Bioinformatics*, 26(8):1112–1118, 2010. doi: 10.1093/bioinformatics/btq099. URL <https://doi.org/10.1093/bioinformatics/btq099>.
- [15] Marija Milacic, Deidre Beavers, Patrick Conley, Chuqiao Gong, Marc Gillespie, Johannes Griss, Robin Haw, Bijay Jassal, Lisa Matthews, Bruce May, et al. The reactome pathway knowledgebase 2024. *Nucleic Acids Research*, 52(D1):D672–D678, 2024. doi: 10.1093/nar/gkad1025. URL <https://doi.org/10.1093/nar/gkad1025>.
- [16] D. N. Nicholson and Casey S. Greene. Constructing knowledge graphs and their biomedical applications. *Computational and Structural Biotechnology Journal*, 18:1414–1428, 2020. doi: 10.1016/j.csbj.2020.05.017. URL <https://doi.org/10.1016/j.csbj.2020.05.017>.
- [17] Dan Ofer, Michal Linial, and Dafna Shahaf. Interfeat: a pipeline for finding interesting scientific features. *Scientific Reports*, 2026. doi: 10.1038/s41598-026-43169-5. URL <https://doi.org/10.1038/s41598-026-43169-5>.
- [18] Open Targets Platform. Open targets platform 22.02 has been released. <https://blog.opentargets.org/open-targets-platform-22-02-release/>, 2022. Accessed 2026-04-26.
- [19] Open Targets Platform. Open targets platform 25.06 has been released. <https://blog.opentargets.org/open-targets-platform-25-06-release/>, 2025. Accessed 2026-04-26.
- [20] Open Targets Platform. Open targets platform data downloads. <https://platform.opentargets.org/downloads>, 2025. Accessed 2026-04-26.
- [21] Florin Ratajczak, Mitchell Joblin, Martin Ringsquandl, and Marcel Hildebrandt. Task-driven knowledge graph filtering improves prioritizing drugs for repurposing. *BMC Bioinformatics*, 23:84, 2022. doi: 10.1186/s12859-022-04608-y.
- [22] Reese A. K. Richardson, Heliodoro Tejedor Navarro, Luis A. Nunes Amaral, and Thomas Stoeger. Meta-research: understudied genes are lost in a leaky pipeline between genome-wide assays and reporting of results. *bioRxiv*, 2023. doi: 10.1101/2023.02.28.530483.
- [23] The Gene Ontology Consortium. The gene ontology knowledgebase in 2023. *Genetics*, 224(1):iyad031, 2023. doi: 10.1093/genetics/iyad031. URL <https://doi.org/10.1093/genetics/iyad031>.
- [24] UniProt Consortium. Uniprot: the universal protein knowledgebase in 2025. *Nucleic Acids Research*, 53(D1):D609–D617, 2025. doi: 10.1093/nar/gkae1010. URL <https://doi.org/10.1093/nar/gkae1010>.
- [25] Hongwei Wang, Fuzheng Zhang, Mengdi Zhang, Jure Leskovec, Miao Zhao, Wenjie Li, and Zhongyuan Wang. Knowledge-aware graph neural networks with label smoothness regularization for recommender systems. *arXiv preprint arXiv:1905.04413*, 2019.

Supplementary Material

S1. Full candidate predictions. The complete set of ~282,500 novel disease–target candidates (score>0.65) for 4,346 diseases, with InterFeat re-ranking features, is released as Outputs/S1-DL_novel_predictions.csv in the anonymized repository.

394 **S2. Top novel and known candidates.** Outputs/S2-DL_novel+known_candidates.csv con-
 395 tains the top-50 novel and known disease–target candidates per disease, intended for direct curatorial
 396 inspection.

397 **S3. Per-feature-group ablation.** A per-feature-group ablation table (text-only, +ontology, +tractabil-
 398 ity, –auxiliary head) is provided in the repository under Outputs/CV_DL/.

399 **S4. Baseline reproduction.** Disease Mean, Target Mean, Open Targets score, Matrix Factorization,
 400 and TF-IDF baselines can be reproduced via `baselines/run_baselines.py`; per-fold metrics are
 401 in Outputs/CV_baselines/.

402 NeurIPS Paper Checklist

403 1. Claims

404 Question: Do the main claims made in the abstract and introduction accurately reflect the
 405 paper’s contributions and scope?

406 Answer: [\[Yes\]](#)

407 Justification: The abstract and Introduction state the contributions (two-tower recommender,
 408 prospective temporal evaluation, repurposing candidate set, interactive platform) that are
 409 quantitatively supported by Tables 1–3 and §4.

410 2. Limitations

411 Question: Does the paper discuss the limitations of the work performed by the authors?

412 Answer: [\[Yes\]](#)

413 Justification: A dedicated §5 discusses auxiliary-head leakage, the single temporal-split run,
 414 illustrative (not systematically validated) discovery examples, the BoW text encoder, the
 415 implicit-feedback PU formulation, and the druggable-genome inference scope.

416 3. Theory assumptions and proofs

417 Question: For each theoretical result, does the paper provide the full set of assumptions and
 418 a complete (and correct) proof?

419 Answer: [\[N/A\]](#)

420 Justification: The paper is empirical and contains no theoretical results.

421 4. Experimental result reproducibility

422 Question: Does the paper fully disclose all the information needed to reproduce the main
 423 experimental results?

424 Answer: [\[Yes\]](#)

425 Justification: Architecture, optimizer, learning rate, batch size, dropout range, validation
 426 fraction, splits, baselines, and OTP release versions are reported in §3; full code and trained
 427 weights are released (Availability).

428 5. Open access to data and code

429 Question: Does the paper provide open access to the data and code, with sufficient instruc-
 430 tions to faithfully reproduce the main experimental results?

431 Answer: [\[Yes\]](#)

432 Justification: Code, preprocessed data, model weights, and ranked predictions are released
 433 under MIT at the repository linked in the Availability section; OTP source data is publicly
 434 available under CC0.

435 6. Experimental setting/details

436 Question: Does the paper specify all the training and test details necessary to understand the
 437 results?

438 Answer: [\[Yes\]](#)

439 Justification: Splits (5×5 target-disjoint CV; 2022→2025 temporal split with leakage filter),
 440 filtering criteria, optimizer (Adam, LR 8×10^{-3} , batch 1,024), early stopping, and baseline
 441 configurations are specified in §3 and §3.3.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about statistical significance?

Answer: [Yes]

Justification: Cross-validation results report mean and standard deviation across 25 target-disjoint folds (Table 1); enrichment statistics are Bonferroni-corrected for the six tests performed (§4.5). The temporal split is a single run; this limitation is explicitly stated in §5.

8. Experiments compute resources

Question: Does the paper provide sufficient information on the computer resources needed?

Answer: [Yes]

Justification: Each cross-validation fold trains in under 30 min on a single consumer GPU (NVIDIA RTX-class, 16 GB); the full 5×5 CV completes overnight. Inference over the $\sim 4,600$ druggable-genome targets $\times \sim 19,000$ diseases ($\sim 87M$ pairs) runs in a few hours on the same hardware. CPU-only training is feasible but slower.

9. Code of ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The work uses only public, properly licensed biomedical resources (Open Targets, EFO, Reactome, UniProt, gnomAD); no human-subject data or personally identifiable information is involved.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts?

Answer: [Yes]

Justification: Positive impacts (orphan-disease candidate generation, reduction of streetlight bias) and risks (mis-prioritization in resource-limited settings, automation bias) are discussed in §5 and the Conclusion, which explicitly frames OTRec as a screening tool that does not replace experimental validation.

11. Safeguards

Question: Does the paper describe safeguards for responsible release of high-misuse-risk data or models?

Answer: [N/A]

Justification: The released model is a target-prioritization recommender over public biomedical data; it is not a generative model, weapons-relevant capability, or content-generation system, and poses no identified high-misuse risk.

12. Licenses for existing assets

Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Open Targets (CC0), EFO (Apache 2.0), gnomAD (ODbL), Reactome (CC-BY 4.0), UniProt (CC-BY 4.0), TensorFlow/Keras (Apache 2.0), CatBoost (Apache 2.0), and XGBoost (Apache 2.0) are all cited; the released code uses MIT.

13. New assets

Question: Are new assets introduced in the paper well documented?

Answer: [Yes]

Justification: The released model weights, training scripts, and ranked candidate tables are documented in the repository README (architecture, intended use, training procedure, license, known limitations).

14. Crowdsourcing and research with human subjects

491 Question: For crowdsourcing experiments and research with human subjects, does the paper
492 include the full text of instructions given to participants, screenshots, and details about
493 compensation?
494 Answer: [N/A]
495 Justification: The work involves no crowdsourcing or human-subject research.

496 **15. Institutional review board (IRB) approvals or equivalent**

497 Question: Does the paper describe potential risks incurred by study participants and IRB
498 approvals?
499 Answer: [N/A]
500 Justification: The work involves no human-subject research and uses only public, de-
501 identified biomedical knowledge bases.

502 **16. Declaration of LLM usage**

503 Question: Does the paper describe the usage of LLMs if it is an important, original, or
504 non-standard component of the core methods?
505 Answer: [N/A]
506 Justification: No LLM is part of the OTRec method; LLMs were used only for incidental
507 writing/editing assistance, which the NeurIPS policy does not require declaring.